Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
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Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

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Session	**2026-27 (ODD)**	**Course Name**	**Lab: Hybrid Computing Systems**
Semester	**7**	**Course Code**	**23IOT1523**
Roll No	**61**	**Name of Student**	**Sheikh Wasimuddin**

Practical Number	**09**
Course Outcome	1. Understand Cloud Computing architecture, service models, deployment models, and virtualization. 2. Apply cloud platforms, storage services, and containerization for application deployment. 3. Implement Edge and Fog Computing solutions for IoT applications. 4. Analyze and develop integrated Cloud-Fog-Edge computing applications based on performance evaluation.
Aim	Performance Comparison of Cloud, Fog, and Edge Computing based on Latency, Response Time, and Resource Utilization.
Theory (100 words)	Cloud, Fog, and Edge Computing are important computing approaches used to manage and process data produced by IoT devices and connected systems. They differ in the location where computation takes place, the distance data travels, and the time required to generate results. These models help improve the efficiency, reliability, and performance of modern applications. **1. Cloud Computing** Cloud computing uses internet-based remote servers to store, manage, and process large amounts of data. IoT devices transmit collected information to cloud data centers, where powerful computing resources perform the required operations. The processed results are then sent back to users or devices. Advantages: • Provides large storage capacity and processing power. • Supports applications with large amounts of data. • Offers flexible and scalable resources.

Limitation: Internet dependency and long-distance data transmission can increase response time.

2. Fog Computing

Fog computing extends cloud services closer to IoT devices by using intermediate computing nodes, such as gateways, routers, and local servers. These nodes analyze data near its source and forward only necessary information to the cloud. This approach improves communication efficiency and reduces network traffic.

Advantages:

- Provides faster processing than distant cloud servers in many situations.
- Reduces bandwidth consumption.
- Supports distributed data processing.
- Improves the responsiveness of IoT applications.

Limitation: Fog nodes require proper configuration, management, and security.

3. Edge Computing

Edge computing processes data directly on or near the devices that generate it. Instead of sending every reading to a remote server, an edge device can analyze sensor information and take immediate action. This model is suitable for applications that need quick decisions and continuous operation.

Advantages:

- Minimizes communication delays.
- Enables rapid responses to changing conditions.
- Reduces unnecessary data transmission.
- Supports real-time monitoring and control.

Limitation: Edge devices may have limited processing power, memory, and storage.

Performance Parameters

	The three computing models can be evaluated using latency, response time, CPU utilization, and memory utilization. Latency measures the delay in data communication, while response time measures the total time taken to complete an operation. CPU and memory utilization indicate the computing resources used during processing. These parameters help compare system performance under different workloads.
Procedure and Execution (100 Words)	Implementation Steps: • Install Python 3.x on the computer and verify the installation using Command Prompt. • Install the required Python libraries, `psutil` and `matplotlib`, using the pip command. • Create a Python program to compare the performance of Cloud, Fog, and Edge Computing. • Generate a common dataset containing sample IoT sensor readings for all three computing models. • Define a data-processing function to perform calculations on the generated dataset. • Simulate Cloud Computing using a higher network delay, Fog Computing using a moderate delay, and Edge Computing using a lower delay. • Use the `time` module to measure latency and total response time during program execution. • Use the `psutil` library to measure CPU and memory utilization. • Execute each computing model five times using the same dataset and calculate the average performance values. • Display the results in the terminal and use Matplotlib to generate graphs comparing latency, response time, CPU usage, and memory usage.

CODE:

```
import time
import random
import statistics
import psutil

# Generate sample sensor data
sensor_data = [random.randint(10, 90) for _ in range(50000)]

def analyze_data(values):
    total = 0
    for item in values:
        total += item * item
    return total

def evaluate_system(system_name, network_delay):
    latency_list = []
    response_list = []
    cpu_list = []
    memory_list = []

    for _ in range(5):
        start_time = time.perf_counter()

        time.sleep(network_delay)

        latency = network_delay * 1000
        cpu_start = psutil.cpu_percent(interval=0.1)
        memory_start = psutil.virtual_memory().percent

        analyze_data(sensor_data)

        cpu_end = psutil.cpu_percent(interval=0.1)
        memory_end = psutil.virtual_memory().percent

        finish_time = time.perf_counter()

        response = (finish_time - start_time) * 1000
        cpu_average = (cpu_start + cpu_end) / 2
        memory_max = max(memory_start, memory_end)

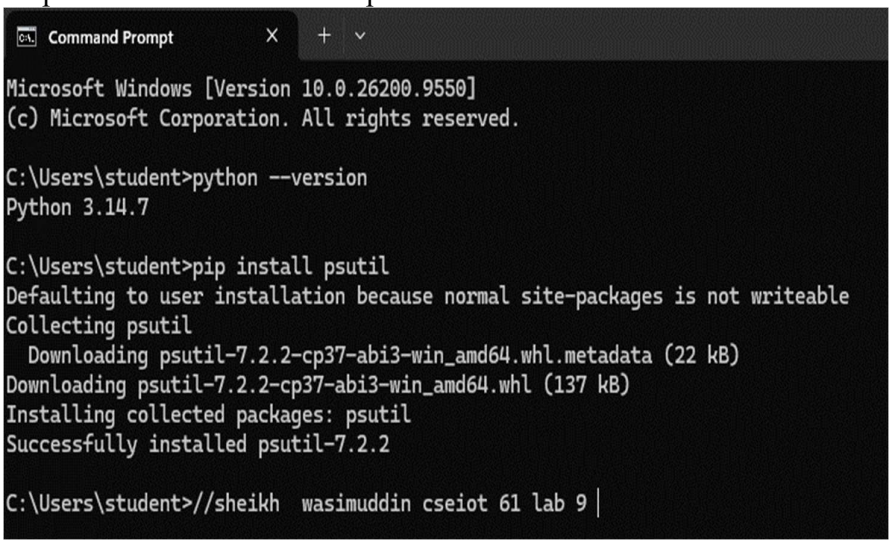
        latency_list.append(latency)
        response_list.append(response)
```

```
        cpu_list.append(cpu_average)
        memory_list.append(memory_max)

    print("\nSystem:", system_name)
    print("=" * 40)
    print("Average Latency:",
          round(statistics.mean(latency_list), 2), "ms")
    print("Average Response Time:",
          round(statistics.mean(response_list), 2), "ms")
    print("Average CPU Usage:",
          round(statistics.mean(cpu_list), 2), "%")
    print("Average Memory Usage:",
          round(statistics.mean(memory_list), 2), "%")

# Compare computing systems
evaluate_system("Cloud Computing", 0.120)
evaluate_system("Fog Computing", 0.060)
evaluate_system("Edge Computing", 0.010)
```

Stepwise Screenshots with steps:



```
import time
import random
import statistics
import psutil

# Generate sample sensor data
sensor_data = [random.randint(10, 90) for _ in range(50000)]

def analyze_data(values):
    total = 0
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        total += item * item
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    latency_list = []
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    memory_list = []

    for _ in range(5):
        start_time = time.perf_counter()

        time.sleep(network_delay)

        latency = network_delay * 1000
        cpu_start = psutil.cpu_percent(interval=0.1)
        memory_start = psutil.virtual_memory().percent

        analyze_data(sensor_data)

        cpu_end = psutil.cpu_percent(interval=0.1)
        memory_end = psutil.virtual_memory().percent

        finish_time = time.perf_counter()

        response = (finish_time - start_time) * 1000
        cpu_average = (cpu_start + cpu_end) / 2
        memory_max = max(memory_start, memory_end)

        latency_list.append(latency)
        response_list.append(response)
        cpu_list.append(cpu_average)
        memory_list.append(memory_max)

    print("\nSystem:", system_name)
    print("=" * 40)
    print("Average Latency:",
          round(statistics.mean(latency_list), 2), "ms")
    print("Average Response Time:",
          round(statistics.mean(response_list), 2), "ms")
```

```
        latency = network_delay * 1000
        cpu_start = psutil.cpu_percent(interval=0.1)
        memory_start = psutil.virtual_memory().percent

        analyze_data(sensor_data)

        cpu_end = psutil.cpu_percent(interval=0.1)
        memory_end = psutil.virtual_memory().percent

        finish_time = time.perf_counter()

        response = (finish_time - start_time) * 1000
        cpu_average = (cpu_start + cpu_end) / 2
        memory_max = max(memory_start, memory_end)

        latency_list.append(latency)
        response_list.append(response)
        cpu_list.append(cpu_average)
        memory_list.append(memory_max)

    print("\nSystem:", system_name)
    print("=" * 40)
    print("Average Latency:",
          round(statistics.mean(latency_list), 2), "ms")
    print("Average Response Time:",
          round(statistics.mean(response_list), 2), "ms")
    print("Average CPU Usage:",
          round(statistics.mean(cpu_list), 2), "%")
    print("Average Memory Usage:",
          round(statistics.mean(memory_list), 2), "%")

# Compare computing systems
evaluate_system("Cloud Computing", 0.120)
evaluate_system("Fog Computing", 0.060)
evaluate_system("Edge Computing", 0.010)
```

```
Command Prompt
Collecting psutil
  Downloading psutil-7.2.2-cp37-abi3-win_amd64.whl.metadata (22 kB)
Downloading psutil-7.2.2-cp37-abi3-win_amd64.whl (137 kB)
Installing collected packages: psutil
Successfully installed psutil-7.2.2

C:\Users\student>python computing_comparison.py

System: Cloud Computing
========================================
Average Latency: 120.0 ms
Average Response Time: 322.81 ms
Average CPU Usage: 0.95 %
Average Memory Usage: 42.1 %

System: Fog Computing
========================================
Average Latency: 60.0 ms
Average Response Time: 263.06 ms
Average CPU Usage: 0.58 %
Average Memory Usage: 42.1 %

System: Edge Computing
========================================
Average Latency: 10.0 ms
Average Response Time: 212.59 ms
Average CPU Usage: 1.04 %
Average Memory Usage: 42.1 %

C:\Users\student>
```

Conclusion	Thus, the performance comparison of Cloud, Fog, and Edge Computing was successfully simulated using Python. Latency, response time, CPU utilization, and memory utilization were measured, and graphs were generated to compare the three computing models.
Date	25/09/2026